

PROBABILITY THEORY AND RANDOM PROCESSES(MA225)

LECTURE SLIDES

Lecture 11

Jointly Distributed Random Variables

Def: A function $\mathbf{X} : \mathcal{S} \rightarrow \mathbb{R}^n$ is called a random vector.

Def: For any random vector $\mathbf{X} = (X_1, X_2, \dots, X_n)$, the joint cumulative distribution function (JCDF) is defined by

$$F_{\mathbf{X}}(x_1, \dots, x_n) = P(X_1 \leq x_1, \dots, X_n \leq x_n),$$

for all $(x_1, \dots, x_n) \in \mathbb{R}^n$.

Remark: $F_X(x) = \lim_{y \rightarrow \infty} F_{X,Y}(x, y)$.

Remark: $F_Y(y) = \lim_{x \rightarrow \infty} F_{X,Y}(x, y)$.

Properties of JCDF

- ① $\lim_{x \rightarrow \infty} \lim_{y \rightarrow \infty} F_{X,Y}(x, y) = 1.$
- ② $\lim_{x \rightarrow -\infty} F_{X,Y}(x, y) = 0$ for all $y \in \mathbb{R}.$
- ③ $\lim_{y \rightarrow -\infty} F_{X,Y}(x, y) = 0$ for all $x \in \mathbb{R}.$
- ④ $F_{X,Y}(\cdot, \cdot)$ is right continuous in each argument keeping other fixed.
- ⑤ For $-\infty < a_1 < b_1 < \infty$ and $-\infty < a_2 < b_2 < \infty,$

$$F_{X,Y}(b_1, b_2) - F_{X,Y}(b_1, a_2) - F_{X,Y}(a_1, b_2) + F_{X,Y}(a_1, a_2) \geq 0.$$

Theorem: Let $G : \mathbb{R}^2 \rightarrow \mathbb{R}$ be a function satisfying above properties. Then G is a JCDF of some 2-dimensional random vector.

Discrete Random Vector

Def: A random vector (X, Y) is said to have a discrete distribution if there exists an atmost countable set $S_{X,Y} \in \mathbb{R}^2$ such that $P((X, Y) = (x, y)) > 0$ for all $(x, y) \in S_{X,Y}$ and $P((X, Y) \in S_{X,Y}) = 1$. $S_{X,Y}$ is called the support of (X, Y) .

Def: Define a function $f_{X,Y} : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f_{X,Y}(x, y) = \begin{cases} P(X = x, Y = y) & \text{if } (x, y) \in S_{X,Y} \\ 0 & \text{otherwise.} \end{cases}$$

The function $f_{X,Y}$ is called joint probability mass function (JPMF) of the DRV (X, Y) .

Properties of JPMF

① $f_{X,Y}(x, y) \geq 0$ for $(x, y) \in \mathbb{R}^2$.

② $\sum_{(x,y) \in S_{X,Y}} f_{X,Y}(x, y) = 1$.

③ $f_X(x) = \sum_{(x,y) \in S_{X,Y}} f_{X,Y}(x, y)$ for fixed $x \in \mathbb{R}$.

f_X is called marginal PMF of X in this context.

Theorem: If a function $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ satisfy 1 and 2 above for the atmost countable set $D = \{(x, y) \in \mathbb{R}^2 : g(x, y) > 0\}$ in place of $S_{X,Y}$, then g is JPMF of some 2-dimensional DRV.

Expectation of Function of DRV

Def: Let (X, Y) be a DRV with JPMF $f_{X,Y}$ and support $S_{X,Y}$. Let $h : \mathbb{R}^2 \rightarrow \mathbb{R}$. Then the expectation of $h(X, Y)$ is defined by

$$E(h(X, Y)) = \sum_{(x,y) \in S_{X,Y}} h(x, y) f_{X,Y}(x, y),$$

provided $\sum_{(x,y) \in S_{X,Y}} |h(x, y)| f_{X,Y}(x, y) < \infty$.

Continuous Random Vector

Def: A random vector (X, Y) is said to have a continuous distribution if there exists a non-negative integrable function $f_{X,Y} : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that

$$F_{X,Y}(x, y) = \int_{-\infty}^x \int_{-\infty}^y f_{X,Y}(t, s) ds dt$$

for all $(x, y) \in \mathbb{R}^2$.

Def: The function $f_{X,Y}$ is called the joint probability density function (JPDF) of (X, Y) .

Def: The set $S_{X,Y} = \{(x, y) \in \mathbb{R}^2 : f_{X,Y}(x, y) > 0\}$ is called the support of (X, Y) .

Properties of JPDP

① $f_{X,Y}(x, y) \geq 0$ for $(x, y) \in \mathbb{R}^2$.

② $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f_{X,Y}(x, y) dx dy = 1$.

③ $f_X(x) = \int_{-\infty}^{\infty} f_{X,Y}(x, y) dy$ for fixed $x \in \mathbb{R}$.

f_X is called marginal PDF of X in this context.

Theorem: If a function $g : \mathbb{R}^2 \rightarrow \mathbb{R}$ satisfy 1 and 2 above, then g is JPDP of some 2-dimensional CRV.

Expectation of Function of CRV

Def: Let (X, Y) be a CRV with JPDP $f_{X,Y}$. Let $h : \mathbb{R}^2 \rightarrow \mathbb{R}$. Then the expectation of $h(X, Y)$ is defined by

$$E(h(X, Y)) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} h(x, y) f_{X,Y}(x, y) dx dy,$$

provided $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |h(x, y)| f_{X,Y}(x, y) dx dy < \infty$.

Examples

Example 1: Let (X, Y) be a DRV with JPMF

$$f_{X,Y}(x, y) = \begin{cases} cy & \text{if } x = 1, 2, \dots, n; y = 1, 2, \dots, n \\ 0 & \text{otherwise.} \end{cases}$$

Example 2: Let (X, Y) be a DRV with JPMF

$$f_{X,Y}(x, y) = \begin{cases} cy & \text{if } x = 1, 2, \dots, n; y = 1, 2, \dots, n; x \leq y \\ 0 & \text{otherwise.} \end{cases}$$

Example 3: Let (X, Y) be a CRV with JPDP

$$f_{X,Y}(x, y) = \begin{cases} ce^{-(2x+3y)} & \text{if } 0 < x < y < \infty \\ 0 & \text{otherwise.} \end{cases}$$